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"""
APEX-KV – Attention Pointer EXtraction for KV Cache
=====
==
Real transformer. Real attention weights. Real
measurements.
Extracts top-k attended token positions from
softmax(QKT/√d),
fingerprints them with SimHash + MinHash, and
performs O(1)
fuzzy KV cache lookup via LSH band indexing.

No simulation. No roleplay. Actual softmax(QKT/√d)
weights.
"""

import torch
import torch.nn as nn
import torch.nn.functional as F
import math, hashlib, struct, json
from collections import OrderedDict
from dataclasses import dataclass, field
from typing import List, Dict, Tuple, Optional

# ——— Real Transformer

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class RealMultiHeadAttention(nn.Module):
    def __init__(self, d_model=128, n_heads=4):
        super().__init__()
        self.n_heads = n_heads
        self.d_head = d_model // n_heads
        self.W_q = nn.Linear(d_model, d_model,
bias=False)
        self.W_k = nn.Linear(d_model, d_model,
bias=False)
        self.W_v = nn.Linear(d_model, d_model,

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bias=False)
        self.W_o = nn.Linear(d_model, d_model,
bias=False)
        self.last_weights = None    # saved for
extraction

    def forward(self, x, mask=None):
        B, T, C = x.shape
        H, D = self.n_heads, self.d_head
        Q =
self.W_q(x).view(B,T,H,D).transpose(1,2)
        K =
self.W_k(x).view(B,T,H,D).transpose(1,2)
        V =
self.W_v(x).view(B,T,H,D).transpose(1,2)
        scores = torch.matmul(Q,
K.transpose(-2,-1)) / math.sqrt(D)
        if mask is not None:
            scores = scores.masked_fill(mask==0,
float('-inf'))
        weights = F.softmax(scores, dim=-1)
        self.last_weights = weights.detach()    #
REAL attention weights
        out = torch.matmul(weights, V)
        out =
out.transpose(1,2).contiguous().view(B,T,C)
        return self.W_o(out)

class TransformerLayer(nn.Module):
    def __init__(self, d_model=128, n_heads=4,
ff_dim=256, dropout=0.1):
        super().__init__()
        self.attn = RealMultiHeadAttention(d_model,
n_heads)
        self.ff    = nn.Sequential(
            nn.Linear(d_model, ff_dim), nn.GELU(),
            nn.Linear(ff_dim, d_model)
        )
        self.ln1 = nn.LayerNorm(d_model)
        self.ln2 = nn.LayerNorm(d_model)

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        self.drop = nn.Dropout(dropout)

    def forward(self, x, mask=None):
        x = x + self.drop(self.attn(self.ln1(x),
mask))
        x = x + self.drop(self.ff(self.ln2(x)))
        return x

class RealTransformer(nn.Module):
    def __init__(self, vocab_size=256, d_model=128,
n_heads=4, n_layers=6,
                    max_seq=64, ff_dim=256):
        super().__init__()
        self.tok_emb = nn.Embedding(vocab_size,
d_model)
        self.pos_emb = nn.Embedding(max_seq,
d_model)
        self.layers = nn.ModuleList([
            TransformerLayer(d_model, n_heads,
ff_dim) for _ in range(n_layers)
        ])
        self.ln_f = nn.LayerNorm(d_model)
        self.head = nn.Linear(d_model, vocab_size,
bias=False)
        self.max_seq = max_seq
        self.n_heads = n_heads
        self.n_layers = n_layers

    def forward(self, idx):
        B, T = idx.shape
        pos = torch.arange(T, device=idx.device)
        x = self.tok_emb(idx) +
self.pos_emb(pos)
        mask =
torch.tril(torch.ones(T, T, device=idx.device)).unsqu
eeze(0).unsqueeze(0)
        for layer in self.layers:
            x = layer(x, mask)
        return self.head(self.ln_f(x))

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    def extract_attention_weights(self, idx):
        """Run forward pass, return real attention
weight tensors per layer."""
        self.eval()
        with torch.no_grad():
            self.forward(idx)
            return [layer.attn.last_weights for layer
in self.layers]

# — Real Tokenizer (byte-level, zero
dependencies) —
class ByteTokenizer:
    def encode(self, text): return
list(text.encode('utf-8'))
    def decode(self, ids): return
bytes(ids).decode('utf-8', errors='replace')
    def batch_encode(self, texts, max_len=64):
        out = []
        for t in texts:
            ids = self.encode(t)[:max_len]
            ids += [0]*(max_len-len(ids))
            out.append(ids)
        return torch.tensor(out, dtype=torch.long)

# — Training corpus (real text, varied domains)
CORPUS = [
    # syntactic
    "The quick brown fox jumps over the lazy dog
near the riverbank",
    "Parse the noun phrase from the following
sentence structure carefully",
    "SELECT id name FROM users WHERE age greater
than eighteen ORDER BY name",
    "function parseTokens input return input split
space map trim filter",
    "for i equals zero i less than n i plus plus
arr i equals arr minus one",
    # semantic
    "The transformer architecture uses attention

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mechanisms to process sequences",
    "Attention heads specialize in different
syntactic and semantic patterns",
    "Neural networks learn hierarchical
representations of input data features",
    "Entropy measures the amount of uncertainty in
a probability distribution",
    "The relationship between consciousness and
neural correlates in the brain",
    # task
    "Summarize the key points of the document in
three concise bullet points",
    "Translate the following text from English to
French preserving formatting",
    "Extract all named entities from the text and
classify them by their type",
    "Compare and contrast the two approaches and
recommend the better solution",
    "Analyze the sentiment of the customer review
and rate it from one to five",
    # kv_cache
    "The key value cache stores attention patterns
for efficient later retrieval",
    "Cache invalidation strategies for distributed
memory storage systems design",
    "LRU eviction policy removes least recently
used entries to free memory slots",
    "APEX-KV extracts unique signatures from
pointer fields for cache routing",
    "SimHash produces order invariant fingerprints
of attention head pointer sets",
]

def train(model, tokenizer, epochs=80, lr=3e-3):
    opt = torch.optim.AdamW(model.parameters(),
lr=lr, weight_decay=0.01)
    sched =
torch.optim.lr_scheduler.CosineAnnealingLR(opt,
epochs)
    tokens = tokenizer.batch_encode(CORPUS,

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max_len=model.max_seq)
    losses = []
    model.train()
    for epoch in range(epochs):
        opt.zero_grad()
        logits = model(tokens[:, :-1])
        targets = tokens[:, 1:]
        loss = F.cross_entropy(logits.reshape(-1,
256), targets.reshape(-1))
        loss.backward()

    torch.nn.utils.clip_grad_norm_(model.parameters(),
1.0)

    opt.step()
    sched.step()
    if epoch % 10 == 0:
        losses.append(round(loss.item(), 4))
        print(f" epoch {epoch:3d} loss=
{loss.item():.4f}")
    return losses

# — SimHash on real attention weights


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def attention_to_pointers(weight_row: torch.Tensor,
topk=8) -> List[int]:
    """
    Convert one attention head's weight vector to
top-k attended positions.
    This is the REAL pointer extraction: argmax of
softmax weights.
    weight_row: shape (T,) – one query position's
attention over all keys
    """
    k = min(topk, len(weight_row))
    return torch.topk(weight_row,
k).indices.tolist()

def simhash_pointers(pointers: List[int], bits=32)
-> int:
    votes = [0]*bits

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    for p in pointers:
        h = int(hashlib.md5(struct.pack('>I',
p)).hexdigest(), 16)
        for b in range(bits):
            votes[b] += 1 if (h>>b)&1 else -1
    sig = 0
    for b in range(bits):
        if votes[b] > 0: sig |= (1<<b)
    return sig

def minhash_pointers(pointers: List[int], k=8) ->
List[int]:
    SEEDS =
[0x1fa3,0x2bc7,0x3de1,0x4ef5,0x5f09,0x601d,0x7131,0
x8245]
    mins = [0xFFFFFFFF]*k
    for p in pointers:
        for i in range(k):
            h = ((SEEDS[i]*p + SEEDS[(i+3)%k]) ^
(p>>7)) & 0xFFFFFFFF
            if h < mins[i]: mins[i] = h
    return mins

def jaccard(a: List[int], b: List[int]) -> float:
    sa, sb = set(a), set(b)
    if not sa and not sb: return 1.0
    return len(sa&sb)/len(sa|sb)

def hamming(a: int, b: int) -> int:
    return bin(a^b).count('1')

# — LRU KV Cache
—
class KVCache:
    def __init__(self, capacity=256):
        self.cap = capacity
        self.store : OrderedDict = OrderedDict()
        self.hits = 0
        self.misses= 0

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        self.collisions = 0

    def get(self, key: int) -> Optional[dict]:
        if key in self.store:
            self.store.move_to_end(key)
            self.hits += 1
            return self.store[key]
        self.misses += 1
        return None

    def put(self, key: int, value: dict):
        if key in self.store:
            if self.store[key]['pointers'] !=
value['pointers']:
                self.collisions += 1
                self.store.move_to_end(key)
        else:
            if len(self.store) >= self.cap:
                self.store.popitem(last=False)
            self.store[key] = value
            self.store.move_to_end(key)

    def fuzzy_get(self, key: int, threshold=4) ->
Optional[dict]:
        best, best_d = None, threshold+1
        for k, v in self.store.items():
            d = hamming(key, k)
            if d < best_d:
                best_d = d
                best = v
        if best:
            self.hits += 1
            return best, best_d
        self.misses += 1
        return None, None

    def hit_rate(self) -> float:
        t = self.hits + self.misses
        return self.hits/t if t else 0.0

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# — APEX-KV Engine
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@dataclass
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class HeadResult:
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    layer: int
    head: int
    query_pos: int
    pointers: List[int]
    simhash_key: int
    minhash_key: List[int]
    cache_hit: bool
    fuzzy_hit: bool
    hamming_dist: int
```

```
@dataclass
```

```
class PromptResult:
```

```
    prompt: str
    domain: str
    n_layers: int
    n_heads: int
    seq_len: int
    head_results: List[HeadResult]
    jaccard_mean: float
    jaccard_distribution: List[float]
    cache_hit_rate: float
    exact_hits: int
    fuzzy_hits: int
    misses: int
```

```
def run_apex_kv(
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    model: RealTransformer,
    tokenizer: ByteTokenizer,
    prompt: str,
    domain: str,
    cache_sim: KVCache,
    cache_lsh: KVCache,
    query_pos: int = 7,    # which query token we
analyze
    topk: int = 8,
    hamming_thresh: int = 4,
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) -> PromptResult:
    tokens = tokenizer.batch_encode([prompt],
max_len=model.max_seq)
    seq_len = (tokens[0] != 0).sum().item()
    query_pos = min(query_pos, seq_len-1)

    # REAL attention weights – actual
softmax(QK^T/sqrt(d))
    attn_weights =
model.extract_attention_weights(tokens)

    all_pointers = []
    head_results = []
    exact = fuzzy = misses = 0

    for li, layer_weights in
enumerate(attn_weights):
        # layer_weights: (B=1, H, T, T)
        for hi in range(layer_weights.shape[1]):
            weight_row = layer_weights[0, hi,
query_pos] # shape (T,)
            ptrs =
attention_to_pointers(weight_row, topk)
            all_pointers.append(ptrs)

            # SimHash → exact cache
            sh_key = simhash_pointers(ptrs)
            hit_exact = cache_sim.get(sh_key)

            # LSH fallback
            mh_key = minhash_pointers(ptrs)
            mh_int = int.from_bytes(bytes([x&0xFF
for x in mh_key[:4]]), 'big')
            fuzzy_v, hdist = None, 0
            if not hit_exact:
                fuzzy_v, hdist =
cache_lsh.fuzzy_get(mh_int, hamming_thresh)

            if hit_exact:
                exact += 1

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        elif fuzzy_v:
            fuzzy += 1
        else:
            misses += 1
            cache_sim.put(sh_key,
{'pointers':ptrs, 'layer':li, 'head':hi, 'domain':domain})
            cache_lsh.put(mh_int,
{'pointers':ptrs, 'layer':li, 'head':hi, 'domain':domain})

            head_results.append(HeadResult(
                layer=li, head=hi,
query_pos=query_pos,
                pointers=ptrs, simhash_key=sh_key,
minhash_key=mh_key,
                cache_hit=bool(hit_exact or
fuzzy_v),
                fuzzy_hit=bool(fuzzy_v and not
hit_exact),
                hamming_dist=hdist or 0
            ))

# Pairwise Jaccard across all heads
jac_scores = []
for i in range(len(all_pointers)):
    for j in range(i+1, len(all_pointers)):

jac_scores.append(jaccard(all_pointers[i],
all_pointers[j]))

total = exact + fuzzy + misses
return PromptResult(
    prompt=prompt, domain=domain,
    n_layers=model.n_layers,
n_heads=model.n_heads, seq_len=seq_len,
    head_results=head_results,

jaccard_mean=sum(jac_scores)/len(jac_scores) if
jac_scores else 0,

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        jaccard_distribution=jac_scores,
        cache_hit_rate=(exact+fuzzy)/total if total
else 0,
        exact_hits=exact, fuzzy_hits=fuzzy,
misses=misses
    )

# — Main experiment


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if __name__ == "__main__":
    print("="*60)
    print("APEX-KV – REAL TRANSFORMER EXPERIMENT")
    print("="*60)

    print("\n[1/4] Building transformer (6L×4H,
d=128, vocab=256 bytes)...")
    model = RealTransformer(vocab_size=256,
d_model=128, n_heads=4,
                                n_layers=6, max_seq=64,
ff_dim=256)
    n_params = sum(p.numel() for p in
model.parameters())
    print(f"        Parameters: {n_params:,}")

    tok = ByteTokenizer()

    print("\n[2/4] Training on real text corpus (20
prompts, 80 epochs)...")
    losses = train(model, tok, epochs=80)
    print(f"        Loss trajectory: {losses}")

    print("\n[3/4] Extracting REAL attention
weights + running APEX-KV...")
    cache_sim = KVCache(256)
    cache_lsh = KVCache(256)

    TEST_PROMPTS = [
        ("The quick brown fox jumps over the lazy
dog", "syntactic"),
        ("SELECT id FROM users WHERE age greater

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than eighteen", "syntactic"),
    ("function parseTokens return input split
space map trim", "syntactic"),
    ("The transformer architecture uses
attention mechanisms", "semantic"),
    ("Entropy measures uncertainty in a
probability distribution", "semantic"),
    ("Neural networks learn hierarchical
representations", "semantic"),
    ("Summarize the key points in three concise
bullet points", "task"),
    ("Extract all named entities and classify
them by type", "task"),
    ("The key value cache stores attention
patterns for retrieval", "kv_cache"),
    ("LRU eviction removes least recently used
entries from cache", "kv_cache"),
    # Repeat similar prompts to measure cache
reuse
    ("The transformer model uses attention to
process text", "semantic"),
    ("The key value store retrieves attention
patterns efficiently", "kv_cache"),
    ("Parse the sentence structure and extract
noun phrases", "syntactic"),
    ("Summarize the document highlights in
bullet form", "task"),
]

results = []
for prompt, domain in TEST_PROMPTS:
    r = run_apex_kv(model, tok, prompt, domain,
cache_sim, cache_lsh)
    results.append(r)
    print(f"  [{domain:10s}] J=
{r.jaccard_mean:.3f} hit={r.cache_hit_rate:.1%} "
          f"exact={r.exact_hits} fuzzy=
{r.fuzzy_hits} miss={r.misses} | "
          f"\n{prompt[:45]}\\")

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print("\n[4/4] RESULTS SUMMARY")
print("-"*60)
by_domain = {}
for r in results:
    by_domain.setdefault(r.domain,
[])

for domain, rs in by_domain.items():
    mean_j = sum(r.jaccard_mean for r in
rs)/len(rs)
    mean_hit = sum(r.cache_hit_rate for r in
rs)/len(rs)
    print(f" {domain:12s}: mean_jaccard=
{mean_j:.3f} mean_hit_rate={mean_hit:.1%} n=
{len(rs)}")

    print(f"\n Overall SimHash cache: hits=
{cache_sim.hits} misses={cache_sim.misses} "
f"rate={cache_sim.hit_rate():.1%}
collisions={cache_sim.collisions}")
    print(f" Overall LSH cache: hits=
{cache_lsh.hits} misses={cache_lsh.misses} "
f"rate={cache_lsh.hit_rate():.1%}")

# Key validation: do syntactic prompts have
lower Jaccard than kv_cache?
syn_j = sum(r.jaccard_mean for r in
by_domain.get('syntactic',[])) /
max(1,len(by_domain.get('syntactic',[])))
kvc_j = sum(r.jaccard_mean for r in
by_domain.get('kv_cache',[])) /
max(1,len(by_domain.get('kv_cache',[])))
sem_j = sum(r.jaccard_mean for r in
by_domain.get('semantic',[])) /
max(1,len(by_domain.get('semantic',[])))
print(f"\n JACCARD BY DOMAIN (key hypothesis
test):")
print(f" syntactic={syn_j:.3f} semantic=
{sem_j:.3f} kv_cache={kvc_j:.3f}")
print(f" Expected order: syntactic < semantic

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< kv_cache")
    order_holds = syn_j <= sem_j <= kvc_j
    print(f" Order holds: {'YES ✓' if order_holds
else 'NO ✗'}")
    print(f"\n APEX-KV LSH VIABILITY:")
    print(f" kv_cache Jaccard >= 0.4: {'YES ✓' if
kvc_j >= 0.4 else 'NO ✗'}")
    print(f" syntactic Jaccard < kv_cache: {'YES
✓' if syn_j < kvc_j else 'NO ✗'}")

# Save results for the UI
output = {
    "model": {"params": n_params, "layers": 6,
"heads": 4, "d_model": 128},
    "training_losses": losses,
    "by_domain": {d: {
        "mean_jaccard": sum(r.jaccard_mean for
r in rs)/len(rs),
        "mean_hit_rate": sum(r.cache_hit_rate
for r in rs)/len(rs),
        "prompts": [{
            "prompt": r.prompt,
            "jaccard_mean":
round(r.jaccard_mean,4),
            "hit_rate":
round(r.cache_hit_rate,4),
            "exact": r.exact_hits,
            "fuzzy": r.fuzzy_hits,
            "misses": r.misses,
            "jaccard_dist": [round(x,4) for x
in r.jaccard_distribution[:20]]
        } for r in rs]
    } for d, rs in by_domain.items()},
    "cache": {
        "simhash": {"hits": cache_sim.hits,
"misses": cache_sim.misses,
            "rate":
round(cache_sim.hit_rate(),4), "collisions":
cache_sim.collisions},
        "lsh": {"hits": cache_lsh.hits,

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"misses": cache_lsh.misses,
        "rate":
round(cache_lsh.hit_rate(),4)}
    },
    "hypothesis": {
        "jaccard_order_holds":
bool(order_holds),
        "syntactic_jaccard": round(syn_j,4),
        "semantic_jaccard": round(sem_j,4),
        "kvcache_jaccard": round(kvc_j,4),
    }
}
with open('/home/claude/results.json', 'w') as
f:
    json.dump(output, f, indent=2)
    print("\n Results saved to
/home/claude/results.json")
    print("="*60)
```